Chapter 18

Explainability, Fairness, and Regulation

18.1 Introduction

Every applied chapter so far has flagged the same unresolved obligation and deferred it here. Chapter 1 warned that a model too opaque to explain to a risk committee is often not deployable, whatever its statistical performance. Chapter 5 showed that collapsing two defensible valuations into one confident-looking number can hide an economically meaningful disagreement. Chapter 6 noted that regulation limits how opaque a credit or insurance model can be, and that interpretability is itself a criterion for choosing among algorithms.

Chapter 15 observed that a suspicious activity report or an account closure often must be explained to a regulator or a customer, which limits how far a fraud or anti-money-laundering program can lean on its least interpretable models; Chapter 16 made the same point about decisions driven by an attention-based language model, and Chapter 17 about the filters and hidden states of convolutional and recurrent networks. This chapter pays off these deferred obligations, not as an afterthought bolted onto a finished model, but as a technical and regulatory discipline with its own methods, worked examples, and open problems.

The chapter is organized around three questions a financial institution must answer before, during, and after deploying a model: *why* a model produced a given output (interpretable and explainable AI); whether it treats people and groups *fairly* under one or more competing legal and ethical definitions; and whether it keeps working as intended once deployed, monitored, and eventually attacked or gamed (robustness, governance, and responsible AI).

Twelve worked examples run throughout, each reusing a model or dataset from an earlier chapter, because explainability and fairness are properties of a model's behavior on real cases. Permutation feature importance and partial dependence work on Chapter 15's fraud-scoring logistic regression; an exact Shapley-value decomposition and a counterfactual explanation address Chapter 9's declined loan applicant; a locally interpretable surrogate is built around Chapter 6's toy feedforward network. A two-group fairness audit, a calibration-by-group check, and adverse-action reason codes rest on the same credit-scoring model; an adversarial "structuring" perturbation extends Chapter 15's money-laundering vocabulary to model evasion; and a population stability index measures score drift, complementing Chapter 15's precision-drift monitoring. Going deeper, a cost-weighted algorithmic-recourse comparison and an integrated-gradients attribution revisit Chapter 6's network to show where a naive local-gradient shortcut fails and a proper path integral does not.

A case vignette on the 2019 Apple Card gender-bias controversy and its regulatory aftermath ties these methods to a real explainability and fairness dispute, and a second, on Upstart Network's engagement with the CFPB, shows the same concerns handled proactively. A final governance subsection turns from models that score to AI agents that act, and sets out the controls such systems need where frameworks written for scoring models only partly reach them.

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18.2 Why Explanation Matters

An unexplained model is not merely opaque; in lending and insurance it is often illegal.

18.2.1 Why Financial AI Needs a Theory of Explanation

Suppose a bank's model validation team asks a data scientist why the credit model just denied a particular applicant, and the honest answer is "because that is what several hundred decision trees, averaged together, produced." The answer is true, but it satisfies no one: not the applicant, who is legally entitled under fair-lending rules to specific reasons for the denial; not the regulator, who must verify that the model is not proxying for a protected characteristic; and not the risk committee, which cannot sign off on a decision it does not understand.

Two related but distinct words recur throughout this chapter, and casual usage often treats them as interchangeable. **Interpretability** is a property of a model itself: a model is interpretable to the degree that a human can see, from its structure alone and without any additional tool, how its inputs map to its outputs. A linear regression's coefficients, Chapter 6's credit-spread regression among them, are interpretable in this sense: each coefficient answers "how much does the prediction change per unit change in this feature."

Explainability, by contrast, is a property of a method applied *after* a model is built: an explanation method takes any model, interpretable or not, and produces a human-comprehensible account of a specific prediction or of the model's general behavior. A deep neural network's millions of weights admit no direct reading, but the network can still be explained, approximately, by the post-hoc methods this chapter develops. Every explanation method here falls into one of four cells formed by two distinctions, illustrated in Figure 18.1: whether it explains the model's behavior in general (*global*) or a single prediction (*local*), and whether it is built into the model's structure (*intrinsic*) or applied afterward to a fitted model (*post-hoc*).

	Intrinsic	**Post-hoc**
Global	Linear/logistic regression coefficients (Ch. 6, 8); a single decision tree	Permutation importance; partial dependence (Section 18.3.1)
Local	One applicant's own linear score decomposition (additive by construction)	Shapley values; LIME (Sections 18.3.2, 18.3.5)

Figure 18.1: A taxonomy of explanation methods along two axes: global versus local, and intrinsic versus post-hoc

The trade-off this taxonomy makes visible is the central tension of the chapter. Chapter 6 observed that the most flexible models, large ensembles, deep neural networks, and transformers (Chapter 16) among them, are typically the least interpretable, while the most interpretable ones, a linear regression or a shallow decision tree, fit a narrower class of relationships. Post-hoc explanation methods do not eliminate this trade-off; they manage it, producing an approximate, human-readable account of a black-box model without changing the model. The approximation is useful (a bank can keep the gradient-boosted model that best predicts default and still produce an adverse-action notice for a declined applicant), but it is an approximation, and Section 18.3.5's worked example shows both how well it can work near the point it is built around and how it degrades away from it.

18.2.2 Intrinsically Interpretable Models Revisited

Before the post-hoc methods, consider choosing a model that needs no post-hoc explanation at all, the left column of Figure 18.1. Chapter 6 introduced its two workhorses: linear and logistic regression, whose

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coefficients are the explanation, and the single decision tree, whose if-then splits read directly as a chain of rules ("if debt-to-income exceeds 30% and credit history is under two years, predict default"). Chapter 9's credit-scoring model and Chapter 15's fraud-scoring model, both logistic regressions, were built this way on purpose, and Section 18.3.2's exact, closed-form Shapley decomposition depends on it. A bank that accepts a small accuracy cost for this direct interpretability avoids the entire post-hoc apparatus.

Two refinements extend this intrinsic column well beyond a single linear model or tree. A **generalized additive model (GAM)** replaces logistic regression's linear combination $\beta_0 + \sum_j \beta_j x_j$ with a sum of flexible, individually plotted functions, $\beta_0 + \sum_j f_j(x_j)$. Each feature's effect can then be nonlinear (a U-shaped effect of age on default risk, say) while the model stays additive, so each f_j can still be inspected on its own, like a linear coefficient. A **credit scorecard**, the industry-standard format behind Chapter 9's 850-to-300 score transformation, goes the other way. It bins each feature into a few ranges (debt-to-income under 20%, 20-30%, over 30%, for instance), assigns each bin an integer number of points from a table, and sums the points across features. A loan officer, auditor, or regulator can verify the result by hand with addition and table lookups. Both formats give up some predictive power, the trade-off of Section 18.2.1, to stay entirely in the intrinsic column and need none of the machinery of Sections 18.3.1 through 18.3.5. That is why scorecards remain the dominant format in consumer credit underwriting, even though tree ensembles and neural networks routinely beat them on held-out accuracy.

18.3 Explanation Methods

This part works the three standard families numerically on models built earlier in the book.

18.3.1 Global Explanations: Permutation Importance and Partial Dependence

Suppose a risk manager wants a one-sentence answer to which of the fraud model's three features it relies on, before deciding whether to keep buying all three data feeds. A global explanation answers this by describing the model's behavior across its input space, or at least across a representative dataset, rather than for one case.

Permutation feature importance asks an empirical question: how much does the model's performance degrade if one feature's values are shuffled across the dataset, breaking that feature's relationship with the outcome while leaving every other feature and the true labels untouched? A large drop marks a feature the model relies on; no change marks one it could apparently do without, at least on this data.

Recall Chapter 15's fraud-scoring logistic regression, fit to ten transactions with three features (amount z-score, distance from home in hundreds of kilometers, and a late-night indicator) and coefficients $\hat{\beta}_0 \approx -2.57$, $\hat{\beta}_1 \approx 0.83$, $\hat{\beta}_2 \approx 0.15$, $\hat{\beta}_3 \approx 0.82$, achieving 90% accuracy and an F_1 of 0.857 at its 0.5 threshold.

Permuting a feature means reassigning its ten values to the ten transactions in a different order; the simplest way by hand is to reverse the column, so that transaction 1 receives transaction 10's value, transaction 2 receives transaction 9's, and so on, while every other feature and every fraud label stays put. Table 18.1 reports the resulting accuracy and F_1 with each feature permuted in turn.

Feature permuted	Accuracy	Precision	F_1
(baseline, no permutation)	0.900	1.000	0.857
Amount z-score	0.300	0.000	undefined
Distance from home	0.900	1.000	0.857
Late-night indicator	0.900	1.000	0.857

Table 18.1: Permutation feature importance for the fraud-scoring model

Reversing the amount z-score column is catastrophic: accuracy collapses from 90% to 30%, and the three transactions flagged after permutation (5, 8, and 10) are all false positives. None of the four frauds is caught, so precision and recall both fall to zero and F_1 is undefined (conventionally reported as zero). Reversing the distance column or the late-night column, by contrast, changes none of the ten predictions, so accuracy and F_1 match the unpermuted baseline.

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The conclusion is not that distance and late-night timing are irrelevant to fraud (Chapter 15's coefficients show that both raise predicted fraud probability), but that in this ten-transaction sample the amount z-score so dominates the linear combination that no rearrangement of the other two features pushes any transaction across the 0.5 threshold. Permutation importance always carries this caveat: it measures a feature's contribution to performance *on the data at hand*, and a feature that looks unimportant here could prove decisive on a larger sample.

A complementary global tool, **partial dependence**, asks a different question: holding every other feature at its empirical values, how does the model's average prediction change as one feature is swept across a range? Formally, the partial dependence of the fraud model on the amount z-score z is $\text{PD}(z) = \frac{1}{n}\sum_{i=1}^{n} \sigma\big(\hat{\beta}_0 + \hat{\beta}_1 z + \hat{\beta}_2 d_i + \hat{\beta}_3 n_i\big)$, averaging over the ten transactions' actual distance and late-night values while only z varies. Table 18.2 evaluates this at six values of z.

Amount z-score	-1	0	1	2	3	4
Average predicted probability	0.095	0.185	0.320	0.489	0.663	0.807

Table 18.2: Partial dependence of predicted fraud probability on amount z-score

The curve crosses the 0.5 threshold between $z = 2$ and $z = 3$, at approximately $z \approx 2.07$: an otherwise typical transaction (average distance and late-night status across the sample) is predicted fraudulent once its amount is roughly two standard deviations above the cardholder's usual spending. Partial dependence and permutation importance answer different questions and can disagree. A feature can have a smooth, informative partial-dependence curve and still show zero permutation importance on a specific sample, as distance did above, because partial dependence describes the model's functional form while permutation importance describes how much one dataset's performance metric depends on that feature.

18.3.2 Local Explanations: Shapley Values for a Single Decision

A global explanation tells a modeler or regulator how a model behaves in general; it does not answer the question a declined applicant asks, which is why *my* application was denied. A **Shapley value**, borrowed from cooperative game theory (Shapley 1953) and popularized in machine learning as SHAP (SHapley Additive exPlanations), answers it by dividing credit for one prediction's deviation from a baseline among the features that produced it. For a model linear in its inputs, as Chapter 9's credit-scoring logistic regression is in the log-odds, the Shapley value has a clean closed form. The contribution of feature j is

$$\phi_j = \hat{\beta}_j (x_j - \bar{x}_j), \tag{18.1}$$

the feature's coefficient times its deviation from a reference baseline. These contributions sum *exactly* to the difference between the individual's predicted log-odds and the baseline's, a property called efficiency that any valid additive attribution must satisfy.

Recall Chapter 9's credit-scoring model,

$$z = -12.152 + 0.530\,(\text{Debt/Income}) - 0.231\,(\text{Credit History})$$

(coefficients shown rounded; the fitted model itself carries full floating-point precision), applied to an applicant with a 28% debt-to-income ratio and 4 years of credit history. A natural baseline is the mean applicant in Chapter 6's six-applicant reference population (debt-to-income $\bar{x}_1 = 26.33\%$, credit history $\bar{x}_2 = 5.17$ years). Comparing the two,

$$\begin{aligned} \text{Applicant:} &\quad z \approx 1.761,\ \Pr(\text{Default}) \approx 85.3\%,\ \text{score} \approx 381, \\ \text{Baseline:} &\quad z_{\text{baseline}} \approx 0.608,\ \Pr(\text{Default})_{\text{baseline}} \approx 64.7\%,\ \text{score} \approx 494, \end{aligned}$$

the applicant's score well below most lenders' approval cutoffs, and the baseline itself subprime but far less severe.

The Shapley decomposition of the gap between them is

$$\underbrace{1.761}_{z_{\text{applicant}}} = \underbrace{0.608}_{z_{\text{baseline}}} + \underbrace{0.530\,(28 - 26.33)}_{\text{debt-to-income contribution} = 0.883} + \underbrace{(-0.231)\,(4 - 5.17)}_{\text{credit-history contribution} = 0.270},$$

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and $0.608 + 0.883 + 0.270 = 1.761$ up to rounding, verifying efficiency. Both features push predicted risk above the baseline: the applicant's debt-to-income is above the reference average (more leverage) and the credit history is shorter (less seasoning). But debt-to-income's contribution (0.883) is more than three times credit history's (0.270), so debt-to-income is the dominant driver of this applicant's elevated risk, not one factor among equals. Figure 18.2 draws the decomposition as a waterfall chart, the standard presentation of a SHAP explanation.

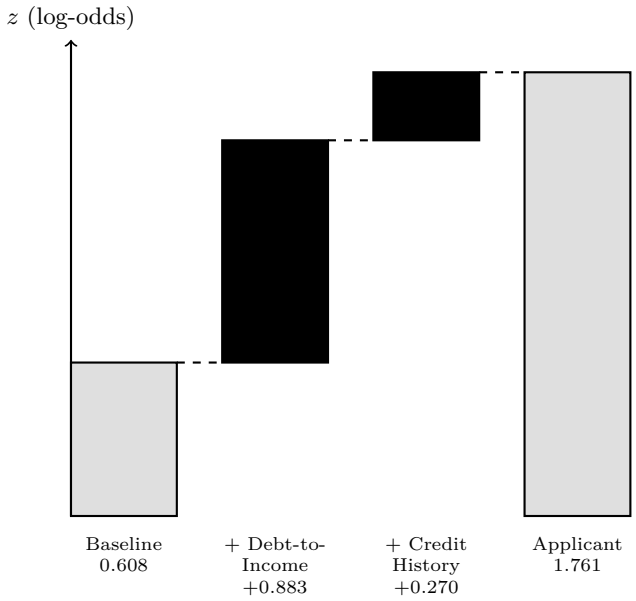


Figure 18.2: The Shapley decomposition as a waterfall chart

18.3.3 From Shapley Values to Adverse Action Reason Codes

This decomposition is, in substance, what U.S. law already requires a lender to produce. The Equal Credit Opportunity Act and its implementing Regulation B require a lender who denies credit to give the applicant the specific, principal reasons for the denial, an *adverse action notice*, rather than a bare rejection. Ranking the Shapley contributions by magnitude produces this: "high debt-to-income ratio relative to similar applicants" as the principal reason (contribution 0.883), and "limited length of credit history" as a secondary reason (contribution 0.270). It works cleanly here because the model is linear in the log-odds. For a nonlinear model such as a gradient-boosted ensemble, the same additive decomposition can still be computed with the general Shapley formula, which averages a feature's marginal contribution over every ordering of the other features and is far more expensive; that is why fast approximations, and simpler surrogates like Section 18.3.5's, remain an active research area.

The Consumer Financial Protection Bureau addressed machine-learning models in two circulars. Circular 2022-03 (CFPB 2022) stated that the adverse-action requirement applies in full to credit decisions based on complex algorithms: a creditor cannot defend a vague notice on the ground that its model is too opaque, so a model whose reasons cannot be extracted is one the creditor should not use. Circular 2023-03 (CFPB 2023) closed the obvious workaround: Regulation B's sample checklist of reasons ("insufficient income," "length of employment") does not suffice when the reason the model relied on, a behavioral or alternative-data feature, is not on the list, and the creditor must state it specifically, even in words of its own. Both circulars were among the 67 guidance documents the Bureau withdrew on 12 May 2025 (CFPB 2025), but a withdrawn interpretation is not a repealed rule: the duty to state the specific, principal reasons sits in the statute and in Regulation B's adverse-action provision, and as of October 2026 neither has changed. The circulars remain the clearest statement of what that duty demands of a black-box model, and so of why the Shapley-to-reason-code mapping above is needed.

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18.3.4 Counterfactual Explanations: How Much Would Need to Change?

A related, often more actionable, explanation answers a different question: not why was I denied, but what is the smallest change that would have gotten me approved? Suppose this lender approves any applicant with a credit score of at least 620, equivalent to

$$\Pr(\text{Default}) \leq \frac{850 - 620}{550} \approx 41.8\%,$$

or $z \leq -0.330$. Holding the applicant's 4 years of credit history fixed and solving

$$-12.152 + 0.530\,(\text{DTI}) - 0.231(4) = -0.330$$

for debt-to-income gives DTI $\approx 24.05\%$, about 3.95 percentage points below the applicant's actual 28%. This *counterfactual explanation* (reduce your debt-to-income ratio by roughly four percentage points, holding everything else fixed, and this model would approve you) is often more useful to a declined applicant than a coefficient or a Shapley value, because it states a concrete target rather than a ranked list of factors. It shares the caveat of Section 18.3.5's local surrogate: it describes what would happen if only debt-to-income changed and nothing else about the applicant shifted with it, a convenient assumption that need not hold in reality.

18.3.5 Local Surrogate Models: LIME for a Genuinely Nonlinear Model

Shapley values have an exact closed form for Chapter 9's credit-scoring model only because it is linear in the log-odds; most flexible models in this book, Chapter 6's ensembles and Chapter 17's CNNs and RNNs, are not. **LIME** (Local Interpretable Model-agnostic Explanations) treats the model as a black box instead: it samples points near the input to be explained, records the black box's predictions there, and fits a simple interpretable model, ordinarily a weighted linear regression, to those input-output pairs. The local model's coefficients are reported as the explanation, valid by construction only in the neighborhood sampled.

Recall Chapter 6's small feedforward network, two inputs, a hidden layer of two ReLU units, and a sigmoid output, with weights

$$W_1 = \begin{pmatrix} 0.4 & -0.2 \\ 0.1 & 0.3 \end{pmatrix},\ b_1 = \begin{pmatrix} 0.05 \\ -0.1 \end{pmatrix},\ W_2 = \begin{pmatrix} 0.6 & -0.5 \end{pmatrix},\ b_2 = 0.1,$$

which at $x_0 = (1.0, 0.5)$ produces hidden activations $h_0 = (0.35, 0.15)$ and a prediction $\hat{y}_0 \approx 0.558$. Both ReLU units are active at x_0 (their pre-activations are positive), so in a small enough neighborhood the network is exactly linear, and the example doubles as a check on LIME: the fitted surrogate should recover the true local behavior almost exactly. Table 18.3 lists eight perturbed points around x_0, each within ± 0.1 of x_0 in each coordinate, with the network's exact prediction at each.

Perturbed input (x_1, x_2)	Hidden activations	Prediction $\hat{y}$
$(1.1, 0.5)$	$(0.39, 0.16)$	0.5632
$(0.9, 0.5)$	$(0.31, 0.14)$	0.5538
$(1.0, 0.6)$	$(0.33, 0.18)$	0.5518
$(1.0, 0.4)$	$(0.37, 0.12)$	0.5651
$(1.1, 0.6)$	$(0.37, 0.19)$	0.5565
$(0.9, 0.4)$	$(0.33, 0.11)$	0.5605
$(1.1, 0.4)$	$(0.41, 0.13)$	0.5698
$(0.9, 0.6)$	$(0.29, 0.17)$	0.5471

Table 18.3: Perturbed samples around $x_0 = (1.0, 0.5)$ and the network's exact prediction

Fitting an ordinary least-squares surrogate

$$\hat{y} \approx a + b_1 x_1 + b_2 x_2 \tag{18.2}$$

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to these eight samples gives $a \approx 0.545$, $b_1 \approx 0.0468$, $b_2 \approx -0.0666$. The analytical local gradient, obtained by differentiating the network directly (valid because both ReLU units stay active throughout this neighborhood), is $\frac{\partial \hat{y}}{\partial x} = \hat{y}_0(1-\hat{y}_0)\, W_2 W_1 \approx (0.0469, -0.0666)$, matching the LIME slope to three decimal places. The surrogate evaluated at x_0 gives $\hat{y}_{\text{surrogate}}(x_0) \approx 0.558$, matching the network's output 0.558: the local fidelity LIME is designed to deliver. The agreement follows from the neighborhood being small enough that no ReLU unit crosses its activation boundary.

Perturbing further away breaks this: at $x = (1.0, -0.1)$, the second hidden unit's pre-activation is exactly zero and the ReLU clips it, giving a true prediction of $\hat{y} \approx 0.594$, while the surrogate extrapolated to the same point predicts approximately 0.598, a discrepancy that widens further out. LIME's neighborhood radius is therefore a modeling choice with real consequences. Too narrow, and the explanation is accurate but says nothing about the model's broader behavior. Too wide, and it averages over a region where the black box is no longer linear, understating the kinked behavior that motivated using a deep model in the first place.

18.4 Regulation, Fairness, and Robustness

Explanation is a legal requirement before it is a technical one, and fairness is measured, not asserted.

18.4.1 International Regulatory Approaches to Explainability

Section 18.3.3 described the U.S. approach to mandated explanation, the adverse-action notice, which is principles-based: it specifies the outcome (specific, principal reasons for a denial) without prescribing a technical method, so Section 18.3.2's Shapley decomposition and Section 18.2.2's summed scorecard points are equally acceptable. The European Union's approach is broader in scope and, in places, more prescriptive. Article 22 of the General Data Protection Regulation (GDPR) restricts decisions based solely on automated processing that produce legal or similarly significant effects, and the Court of Justice of the EU held in 2023 that a credit bureau's score can itself be such a decision when lenders rely on it decisively (CJEU 2023). How much individualized explanation Articles 13–15 then demand, a global description of the model's logic or an account of the specific decision, remains contested among legal scholars, a rare case in this book where the technical machinery of Sections 18.3.2 and 18.3.5 is more settled than the legal question of how much of it is required. The EU's Artificial Intelligence Act (EU 2024) moves from a data-protection framing to a product-safety one: it lists credit scoring and life and health insurance pricing as high-risk systems that must be documented, overseen, and tested *before* deployment, rather than explained one decision at a time afterward.

Table 18.4 sets side by side the regimes a credit or insurance-pricing model is most likely to meet, as of October 2026. Much of it is supervisory guidance rather than statute, and it moves quickly: five of its rows changed during 2026 alone.

The table's most consequential pattern is divergence. Between 2023 and 2026 the EU, the UK, Canada, and Singapore all moved toward more explicit expectations for AI models in credit and insurance, while the U.S. federal government moved the other way on fairness. Executive Order 14281 of April 2025 (Executive Order 14281 2025) directs federal agencies to deprioritize disparate-impact enforcement, and the April 2026 amendment of Regulation B (CFPB 2026) withdrew federal reliance on disparate impact under the Equal Credit Opportunity Act while leaving the adverse-action explanation duty untouched; the government-wide AI-safety agenda of Executive Order 14110 (2023) had already been revoked on 20 January 2025. That does not make fairness testing optional for a U.S. lender. The Fair Housing Act, under which the Supreme Court recognized disparate-impact claims in 2015, state fair-lending and consumer-protection laws (Section 18.4.2), and the regimes abroad still reach outcomes, and a model built for one market is often deployed in several. A firm operating in both the EU and the U.S. therefore holds two compliance postures on the same model: explaining individual outcomes after the fact on one side, and certifying the system's design, documentation, and human oversight before it scores a real applicant on the other. The same toolkit, Sections 18.3.1 through 18.3.5, serves both. The U.S. banking agencies' replacement of SR 11-7 by SR 26-2 (Federal Reserve, OCC, and FDIC 2026) in April 2026 changes less for this chapter than its headline suggests: the new guidance keeps conceptual soundness, ongoing monitoring, and outcomes analysis, makes the intensity of validation explicitly risk-based, and leaves generative and agentic AI out of scope pending separate treatment, so the scoring and fraud models of this chapter remain inside it.

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Jurisdiction	Instrument	What it asks of a credit or pricing model	Status (October 2026)
EU	AI Act (Reg. 2024/1689), general provisions	Prohibited practices (e.g., social scoring); AI literacy; separate obligations for general-purpose AI models	In force 1 Aug 2024; prohibitions from 2 Feb 2025; GPAI obligations from 2 Aug 2025
	AI Act, Annex III, points 5(b)–(c)	Credit scoring of natural persons (fraud detection excluded) and life and health insurance risk assessment and pricing are high-risk: risk management, data governance, technical documentation, logging, human oversight, accuracy and robustness; deployers assess fundamental-rights impact	Originally 2 Aug 2026; the Digital Omnibus (proposed 19 Nov 2025; Reg. 2026/1744, in force 27 Jul 2026) defers it to 2 Dec 2027
US federal	ECOA and Regulation B	Specific principal reasons for every adverse action; no discrimination on a prohibited basis	2026 amendment, effective 21 Jul 2026, removes the disparate-impact "effects test"; disparate treatment still prohibited
	CFPB Circulars 2022-03 and 2023-03	Neither a complex algorithm nor the sample-form checklist excuses vague adverse-action reasons	Issued 2022 and 2023; withdrawn 12 May 2025; the underlying duty is unchanged
	SR 11-7 (2011), now SR 26-2 / OCC Bulletin 2026-13	Model risk management: conceptual soundness, ongoing monitoring, outcomes analysis, independent validation, scaled to risk	SR 26-2 superseded SR 11-7 on 17 Apr 2026; generative and agentic AI out of scope
US states	Colorado SB 24-205, replaced by SB 26-189	Original: duty of care against algorithmic discrimination in consequential decisions (incl. lending, insurance), impact assessments. Replacement: notice, plain-language explanation within 30 days of an adverse decision, data correction, human review	SB 24-205 never took effect (delayed to 30 Jun 2026, then repealed); SB 26-189 signed 14 May 2026, main duties from 1 Jan 2027
	NYDFS Insurance Circular Letter No. 7	Insurers using AI or external consumer data in underwriting and pricing must show it is not unfairly discriminatory, test for disparate impact, govern and document it, and oversee vendors	Issued 11 Jul 2024
UK	PRA SS1/23	Five model risk management principles for banks, explicitly covering AI and machine-learning models	Effective 17 May 2024
	FCA Consumer Duty	Good outcomes for retail customers: fair price and value, avoiding foreseeable harm; reaches algorithmic pricing and decisioning	Open products 31 Jul 2023; closed products 31 Jul 2024
Canada	OSFI Guideline E-23	Enterprise-wide model risk management over the model lifecycle, explicitly including AI and machine-learning models, at federally regulated banks and insurers	Final 11 Sep 2025; effective 1 May 2027
Singapore	MAS FEAT principles, Veritas, and Guidelines on AI Risk Management	FEAT (2018): fairness, ethics, accountability, transparency, with Veritas assessment methods; Guidelines: AI inventory, risk-materiality assessment, lifecycle controls, board oversight	FEAT voluntary; Guidelines issued 7 Oct 2026, effective 7 Oct 2027 (some sections by 7 Oct 2028)
China	PIPL, Art. 24	Transparent, fair automated decisions with no unreasonable differential treatment in prices or terms; right to an explanation and to refuse solely automated decisions with significant effects	Effective 1 Nov 2021
	Algorithmic Recommendation Provisions	Algorithm filing; option to switch off personalization; no unreasonable price discrimination from user profiles	Effective 1 Mar 2022
	Generative AI Interim Measures	Lawful training data, content labeling, security assessment and filing for public-facing generative AI services	Effective 15 Aug 2023

Table 18.4: AI and model-risk regimes relevant to credit and insurance-pricing models, as of October 2026. Compiled from the instruments themselves (EU 2024; EU 2026; CFPB 2026; CFPB 2022; CFPB 2023; CFPB 2025; Federal Reserve, OCC, and FDIC 2026; Executive Order 14281 2025; Colorado General Assembly 2026; NYDFS 2024; PRA 2023; FCA 2022; OSFI 2025; MAS 2026; NPC Standing Committee 2021).

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One further development closes the loop: regulators now run models themselves. Supervisory technology, SupTech, is AI deployed by the supervisor rather than the supervised: securities regulators mine trading and filing data for the manipulation patterns Chapter 14 described, financial-intelligence units apply network analytics to the suspicious-activity reports Chapter 15's institutions file, and central banks apply text methods to their supervisory correspondence to triage which firms need attention. For a model builder, the consequence is symmetry: a bank's AI system is increasingly examined *by* an AI system, and the governance artifacts this chapter argues for (documentation, monitoring records, explanation apparatus) are what that examination consumes. It also brings this chapter's problems to the other side of the table: a supervisor's model can be biased, opaque, or wrong at system-wide scale, which is why supervisory bodies have begun publishing governance standards for their own tools that look strikingly like the SR 11-7 discipline they impose on the firms they oversee.

18.4.2 Fairness in Financial AI

Explainability answers what drove a specific decision; fairness asks a logically separate question, whether a model's decisions, however well explained, treat different groups equitably. Regulators have historically operationalized fairness through two statistical definitions that are not entirely compatible, and theoretical work has shown that the incompatibility is, in general, a mathematical impossibility rather than a matter of insufficiently clever engineering.

Disparate impact and the four-fifths rule. A lending, hiring, or underwriting practice exhibits disparate impact if it produces substantially different outcome rates across groups, even when no feature encodes group membership and no discriminatory intent can be shown. U.S. regulatory guidance operationalizes "substantially different" with the *four-fifths rule*: if the group with the lower favorable-outcome rate receives that outcome less than 80% as often as the group with the higher rate, the *adverse impact ratio*,

$$\text{AIR} = \frac{\min(\text{rate}_A, \text{rate}_B)}{\max(\text{rate}_A, \text{rate}_B)}, \tag{18.3}$$

falls below 0.80, disparate impact is presumed and the practice invites regulatory scrutiny regardless of intent. The rule comes from employment law: it was written into the 1978 Uniform Guidelines on Employee Selection Procedures (EEOC 1978) as a rule of thumb for hiring tests, and fair-lending analysts borrowed it as a screening heuristic, a flag for where to look harder rather than a legal threshold for credit. Its federal footing in lending narrowed in 2026, when the CFPB amended Regulation B to state that the Equal Credit Opportunity Act does not support disparate-impact liability (Section 18.4.1), but the Fair Housing Act, state law, and lenders' own model-risk programs keep the ratio in routine use.

Equalized odds and error-rate parity. A second family of definitions asks whether the model is equally *accurate* for each group: whether its false-negative rate (missing an applicant who will default, in this chapter's running example) and false-positive rate are similar across groups. A model can satisfy the four-fifths rule while its false-negative rate is far higher for one group, or the reverse. A theoretical literature (Chouldechova 2017; Kleinberg et al. 2017), developed for criminal-justice risk scores but directly applicable here, proves something stronger: unless the two groups have identical underlying default rates, no model can simultaneously satisfy calibration, equal false-negative rates, and equal false-positive rates across groups. A regulator or model-risk function must therefore choose explicitly which fairness definition a model is held to, rather than assume that satisfying one satisfies the others.

Table 18.5 makes this tension concrete with a small fairness audit of Chapter 9's credit-scoring model applied to two hypothetical four-applicant groups, Alpha and Beta, at the same 620-score approval threshold used in Section 18.3.4 ($\Pr(\text{Default}) \leq 41.8\%$).

Group Alpha's approval rate is $3/4 = 75\%$; Group Beta's is $1/4 = 25\%$, giving an adverse impact ratio of $0.25/0.75 = 0.333$, far below the 0.80 threshold: on this evidence alone, a clear four-fifths-rule violation calling for further fair-lending investigation. The error rates tell a different story. Of Group Alpha's two future defaulters (Alpha-2 and Alpha-3), the model approves Alpha-2, whose moderate debt-to-income and credit history keep her predicted probability just under the threshold, a false-negative rate of 50%. Of Group Beta's two future defaulters (Beta-2 and Beta-4), the model denies both, a false-negative rate of 0%. The group approved far more often (Alpha) is the group whose risk the model is worse at catching, while the group facing the harsher approval rate (Beta) is, on this small sample, screened perfectly. A regulator focused

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Applicant	Debt/income	Credit history	Pr(Default)	Decision	Will actually default
Alpha-1	15%	9 yrs	0.2%	Approve	No
Alpha-2	22%	6 yrs	13.2%	Approve	**Yes**
Alpha-3	30%	3 yrs	95.5%	Deny	Yes
Alpha-4	18%	8 yrs	1.1%	Approve	No
Beta-1	25%	4 yrs	54.3%	Deny	No
Beta-2	33%	2 yrs	99.2%	Deny	Yes
Beta-3	20%	5 yrs	6.2%	Approve	No
Beta-4	38%	1 yr	>99.9%	Deny	Yes

Table 18.5: A fairness audit of the credit-scoring model on two synthetic four-applicant groups

only on approval-rate parity would flag the model for disadvantaging Beta; one focused only on error-rate parity would note that Alpha's approved population carries more missed risk. Reconciling the two, rather than optimizing either in isolation, is the judgment call that the impossibility result says no purely statistical fix can avoid.

The impossibility result names a third statistic, calibration, worth checking directly. Both groups share an actual default rate, $2/4 = 50\%$ for Alpha and $2/4 = 50\%$ for Beta, the special case the theorem carves out as the one setting where calibration, equal false-negative rates, and equal false-positive rates could in principle coexist.

Yet a calibration-in-the-large check, comparing each group's average *predicted* default probability with its actual default rate, shows the model is not equally calibrated across them:

$$\begin{aligned} \text{Group Alpha predictions:} &\quad 0.19\%,\ 13.26\%,\ 95.50\%,\ 1.14\% \quad (\text{mean } 27.5\%,\ \text{actual } 50\%), \\ \text{Group Beta predictions:} &\quad 54.3\%,\ 99.2\%,\ 6.2\%,\ 99.96\% \quad (\text{mean } 64.9\%,\ \text{actual } 50\%), \end{aligned}$$

the model under-predicts risk for Alpha and over-predicts it for Beta, although both groups' realized outcomes are identical. This does not violate the impossibility theorem, which concerns idealized population distributions rather than any one small sample, but it shows how the tension appears in practice: equal base rates do not force equal calibration, equal error rates, or equal approval rates in a finite population, and an audit that checked only the adverse impact ratio, only the error-rate gap, or only calibration would each tell an incomplete part of the same story.

Individual Fairness: A Third Paradigm

Both definitions above are *group* fairness notions: they compare an aggregate statistic, an approval rate or an error rate, across predefined groups. A third paradigm, **individual fairness** (Dwork et al. 2012), asks whether two individuals who are similar with respect to the task are treated similarly, regardless of group. Formally, this needs a task-specific similarity metric, a distance between applicants that captures what "similar with respect to creditworthiness" means, and a model is individually fair if it maps similar inputs to similar outputs under that metric.

Table 18.5's Alpha-2 and Beta-3 show why this lens matters: their predicted default probabilities, 13.2% and 6.2%, are broadly comparable despite their different groups, and both are approved (though a stricter similarity threshold could flag even this gap as worth investigating). The practical difficulty is the one this parenthetical hints at. Choosing the similarity metric requires the same value judgment that group definitions push onto the choice of protected groups and outcome statistics: which features are legitimately relevant to creditworthiness. Individual fairness therefore relocates the value judgment of Section 18.4.2's impossibility result rather than resolving it, from "which group statistic to equalize" to "which similarity metric to adopt."

Fairness Testing in Practice: Proxies and Less Discriminatory Alternatives

Table 18.5 took group membership as known. In most U.S. consumer lending it is not: Regulation B generally forbids a lender to ask an applicant's race or ethnicity, except for the mortgage data collected under the

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Home Mortgage Disclosure Act, so a fair-lending analyst testing an auto-loan or credit-card model must first estimate it. The standard tool is **Bayesian Improved Surname Geocoding (BISG)** (Elliott et al. 2009), which the CFPB adopted and published for its own fair-lending analysis (CFPB 2014). It is Bayes' rule from Chapter 1: the Census Bureau's surname list supplies a prior probability of each race and ethnicity given the applicant's surname, and the racial composition of the applicant's census block group updates it,

$$\Pr(\text{race} \mid \text{surname}, \text{location}) \propto \Pr(\text{race} \mid \text{surname}) \times \Pr(\text{location} \mid \text{race}),$$

under the assumption that surname and location are independent given race. The output is a probability for each group, not a label, and good practice carries those probabilities through the audit, weighting each applicant's outcome by them, rather than rounding each applicant into one group. Proxy error is not random, since it depends on where people live and how distinctive their surnames are, so a disparity measured through BISG can be biased in either direction (Chen et al. 2019); an audit should report it as an estimate with that caveat.

Finding a disparity starts the analysis rather than ending it. U.S. disparate-impact doctrine runs in three steps: a disparity is shown, the lender shows a legitimate business need for the practice, and the question becomes whether a **less discriminatory alternative** would serve that need about as well. For a machine-learning model the third step is a search problem. Many models fit the same data almost equally well, the phenomenon called model multiplicity (Black et al. 2022), so the analyst retrains over different feature sets, hyperparameters, constraints, and thresholds and looks among the near-equally accurate candidates for the one with the highest adverse impact ratio (Gillis et al. 2024). The CFPB no longer presses that search under the Equal Credit Opportunity Act after 2026, but state enforcers still treat untested disparities as a violation: in July 2025 the Massachusetts Attorney General settled with the student lender Earnest Operations for $2.5 million over algorithmic underwriting alleged to produce disparate outcomes for Black, Hispanic, and non-citizen applicants, and over inaccurate adverse-action notices (Massachusetts Attorney General 2025). The settlement required the kind of model governance and fairness testing this section describes.

18.4.3 Case Vignette: The Apple Card Gender-Bias Controversy

In November 2019, software entrepreneur David Heinemeier Hansson posted that the Apple Card, issued by Goldman Sachs, had given him a credit limit twenty times his wife's, although the couple filed joint tax returns and his wife had a longer credit history and a comparable or better credit score. Apple co-founder Steve Wozniak reported a similar disparity with his own spouse. The claims spread rapidly and prompted the New York State Department of Financial Services to open a formal inquiry into whether Goldman Sachs's credit-limit algorithm violated fair-lending law.

The regulator's findings, published in 2021, illustrate this chapter's distinctions clearly. The investigation found no evidence that the algorithm used sex or gender as an input, and no evidence of intentional discrimination under the Equal Credit Opportunity Act, the claim the controversy had focused on.

It did find that Goldman Sachs's underwriting process lacked the testing and monitoring Section 18.4.2 formalizes. The bank could not demonstrate after the fact that its model did not produce disparate outcomes correlated with sex, because it had not tested for this in the way a four-fifths-rule audit like Table 18.5's would. And its customer service representatives, asked to explain a specific credit-limit decision, could offer no model-derived reason of the kind Section 18.3.3's reason codes produce, only that "the algorithm decided." The regulator recommended not a fine but stronger fairness testing and clearer consumer-facing explanations of algorithmic credit decisions, the two capabilities this chapter builds by hand. The episode corrects a common misreading of fair-lending law. Leaving a protected characteristic out of a model's inputs, sometimes called "fairness through unawareness," is neither necessary nor sufficient to avoid disparate impact. Facially neutral inputs correlated with a protected characteristic, such as a joint account structure reflecting gendered patterns in whose name assets are held, can reproduce the disparity a cruder model would produce by using the characteristic directly.

Postscript (October 2024). The Apple Card came back before a regulator for a different reason. On 23 October 2024 the CFPB ordered Goldman Sachs to pay a $45 million civil penalty and at least $19.8 million in redress to consumers, and Apple a $25 million penalty (CFPB 2024). The orders did not concern credit limits or sex. Apple had failed to send tens of thousands of customers' transaction disputes to Goldman, Goldman had not investigated the disputes it did receive as federal law requires, and both had misled customers about

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interest-free monthly installment plans for Apple devices; Goldman was also barred from launching a new credit card without a credible plan to comply with the law. (The Bureau terminated Apple's order on 22 September 2025, after Apple had paid its penalty.) In 2021 the gap was a model nobody had tested for disparate outcomes or could explain to a customer; in 2024 it was a dispute system launched four days after Goldman's board had been warned it was not fully ready. Neither was a defect in the scoring model's inputs; both were failures of the governance around it, the subject of Section 18.5.1.

18.4.4 Adversarial Robustness in Financial Models

A model's outputs can be manipulated deliberately, not just misunderstood, and Chapter 15's money-laundering vocabulary already has a name for this: *structuring*, splitting a transaction to fall just under a reporting or detection threshold. The idea applies directly to evading a fraud-scoring model rather than a currency-reporting requirement.

Recall Chapter 15's transaction 1, with an amount z-score of 3.1, a distance of 800 kilometers from the cardholder's home, and a late-night flag, correctly flagged as fraud with predicted probability 88.3%. Holding distance and late-night status fixed and solving $-2.57 + 0.83\,z + 0.15(8.0) + 0.82(1) = 0$ for z gives $z \approx 0.663$. A fraudster who could disguise the amount so that it appears only 0.663 standard deviations above the cardholder's typical spending, for instance by splitting it into several smaller purchases at the same distant, late-night location, would push the prediction just under the 0.5 threshold and evade detection. The required change is large, a reduction of 2.44 in the amount z-score, because the transaction's distance and timing already carry so much of the model's suspicion. This is a more reassuring finding than the classic adversarial examples in image classification, where an imperceptibly small pixel-level perturbation (Goodfellow et al. 2015) can flip a deep network's prediction. A low-dimensional model, linear in the log-odds, with several independently suspicious features is much harder to game through any single feature than a high-dimensional deep network. It remains vulnerable, though, to an adversary who manipulates several features jointly (a smaller amount, a nearby location, business hours), an attack this single-feature analysis does not rule out.

This is why Chapter 17's point about vendor black-box processing and Chapter 15's point about adversarial adaptation both matter here. Any model whose scoring logic can be inferred by the population it scores will see its inputs shift in response, whether the inference comes from a fraud ring or from market participants learning which alternative-data patterns a fund's model rewards. That is the concept-drift dynamic Chapter 14 introduced for high-frequency trading models and Chapter 15 quantified for fraud-model precision.

Adversarial robustness concerns an adversary crafting an input to fool the model. It differs from the stress testing of Chapters 8 and 9, which subjects a model or portfolio to an extreme but plausible scenario such as a severe recession or a market crash. Both ask how a model behaves outside the data it was trained on, but stress testing probes a shift in the whole population (unemployment rising across the board, say), while adversarial robustness probes a targeted manipulation of one case, and a model can be robust to one and vulnerable to the other. A fraud model retrained often enough to absorb an economy-wide shift in spending can still be gamed one transaction at a time, as shown above. And a credit model that behaves sensibly in a simulated recession offers no guarantee against the manipulation risk Section 18.3.4 raised: an applicant who learns the threshold and adjusts a self-reported input to cross it, without any change in creditworthiness.

18.5 Governance and Supervisory Engagement

A model in production drifts, and this part covers what is monitored and what is done about it.

18.5.1 Model Governance and Monitoring in the AI Era

Chapters 8 and 9 introduced model risk management, independent validation and ongoing monitoring for market and credit risk models, and Chapter 15 formalized the three-lines-of-defense structure (business unit, independent compliance and risk function, internal audit) for fraud and anti-money-laundering programs. U.S. bank regulators codify this discipline for essentially every model a bank relies on in SR 11-7 (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency 2011) (superseded in April 2026 by SR 26-2, which keeps all three pillars; Section 18.4.1). It names conceptual soundness, ongoing monitoring, and outcomes analysis as the three pillars of sound model risk management, matching the three

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capabilities this chapter builds: understanding why a model produces an output, tracking whether it keeps working, and checking its outcomes for fairness.

Chapter 15's precision-drift table tracked a fraud model's precision falling from 24.2% to 17.1% over six months as fraudsters adapted, but precision is a *lagging* indicator: computing it requires investigators to confirm which flagged transactions were fraudulent, which takes time.

The **population stability index (PSI)** offers a complementary, *leading* indicator that needs no labels, only the model's score outputs: it compares the current period's score distribution with a baseline established when the model was validated. Formally,

$$\text{PSI} = \sum_i (c_i - b_i) \ln(c_i / b_i), \tag{18.4}$$

summed over score bins i with baseline proportion b_i and current proportion c_i. Table 18.6 illustrates this for the fraud-scoring model, comparing its validation-time score distribution across four bins against a later snapshot.

Score bin	$[0, 0.25)$	$[0.25, 0.50)$	$[0.50, 0.75)$	$[0.75, 1.00]$	Total PSI
Baseline proportion	0.50	0.25	0.15	0.10	
Current proportion	0.34	0.29	0.22	0.15	
Bin contribution	0.062	0.006	0.027	0.020	0.115

Table 18.6: Population stability index for the fraud-scoring model's score distribution

A total PSI of 0.115 falls into the conventional "moderate shift, worth investigating" band of credit-risk monitoring (below 0.10 is treated as no material change; above 0.25 as a major shift typically requiring retraining or revalidation), well before precision has moved enough to be distinguishable from noise. A smaller share of transactions now scores in the lowest bin and a larger share in every higher bin: the score population has shifted upward, a sign that the transactions the model sees have changed (a new customer segment, a different merchant mix, an emerging fraud pattern), and PSI reports it before enough investigated cases accumulate to move precision. The adaptive evasion of Section 18.4.4 would look different. Structuring pushes only the fraudulent transactions' scores *down*, and because fraud is a small share of transactions, that shift can hide inside an aggregate distribution whose PSI barely moves, one reason to monitor scores by segment as well as in aggregate. This is the early warning SR 11-7's ongoing-monitoring pillar is meant to produce, and why a mature governance program tracks both a leading, label-free statistic like PSI and a lagging, outcome-based one like precision.

The National Institute of Standards and Technology's AI Risk Management Framework (NIST AI RMF 1.0; NIST 2023) organizes this chapter's concerns (explainability, fairness, robustness, and governance) into a voluntary framework built around four functions: *govern* (accountability and policy: Section 18.5.1 and Chapter 15's three lines of defense), *map* (a model's context and likely impact, including on demographic groups: Section 18.4.2), *measure* (the methods of Sections 18.3.1 through 18.3.5 and 18.4.4), and *manage* (acting on what measurement reveals, from adverse-action notices to model retirement). The framework is deliberately not finance-specific and not a checklist to complete once. It is best read as an organizing structure for a discipline that, as the Apple Card vignette (Section 18.4.3) showed, tends to fail in the gaps between the four functions: an algorithm measured for accuracy but never mapped for demographic impact, or governed on paper with no one able to produce an explanation when a regulator asked for one.

18.5.2 Governing Systems That Act

Every model this chapter has governed so far produces a number (a score, a probability, a flag) that a person or a separate process then acts on, and every tool it has built, from reason codes to the population stability index, governs that number. The LLM agents of Chapter 16's section on agentic AI break this arrangement: they call tools, and the tools change records, send messages, and move money. Chapter 10 made the central point: an AI that acts closes the detection window every downstream control depends on, so the first evidence of a failure is an executed action rather than a declined proposal. This subsection asks what that changes for governance.

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Delegated authority. Banks govern people through delegated authority: what each person may approve, sign, or trade, up to what amount, is written down, granted by someone entitled to grant it, and reviewed. An agent needs the same instrument, a mandate that states its tasks, its tools, the data and counterparties it may touch, and its value thresholds, and names the human who owns it. Two rules follow. An agent's authority can never exceed that of the person delegating to it, and an agent serving several people must not accumulate the union of their entitlements, which would recreate the segregation-of-duties breach Chapter 10 warned an agent holds by construction. FINRA's 2026 oversight report, published in December 2025 and the first to address AI agents directly, lists this among their risks: agents that act beyond the scope and authority the user intended (FINRA 2025).

Accountability for actions rather than predictions. Validating a scoring model asks whether its outputs are accurate, stable, and fair. Validating an agent must also ask, for each action, whether it was authorized, whether it was correct, and who answers for it. The firm does: an agent is not a legal person, and neither the model vendor nor the agent absorbs the firm's obligations to its clients or supervisors. In practice this turns outcomes analysis, SR 11-7's third pillar, into the trajectory review Chapter 16 described: reading samples of what the agent did, step by step, not only what it concluded.

Action logs and audit trails. An action that cannot be reconstructed cannot be reviewed, contested, or corrected. An agent's log must be richer than a model's prediction log: the instruction it received, the context it retrieved, each tool call with its arguments and result, each human approval with the approver's identity, and the versions of model, prompt, and tool set in force. This is the agent counterpart of the EU AI Act's requirement that high-risk systems automatically record events over their lifetime (Article 12) (EU 2024), and of FINRA's expectation that firms track agent actions and decisions (FINRA 2025). The log is also the raw material for the supervision by machine described in Section 18.4.1.

Limits and kill switches. Because an agent acts at machine speed, its failures must be bounded before anyone notices them: caps on the value of any single action, on actions per hour, and on total spending, and a kill switch that halts the agent and leaves its work in a known state. Electronic trading learned this expensively, as Chapter 14's account of Knight Capital showed, and the EU AI Act's human-oversight article requires that an overseer be able to interrupt a high-risk system through a "stop" button or similar procedure that brings it to a halt in a safe state (Article 14(4)(e)) (EU 2024).

Human approval for irreversible actions. Least privilege decides what an agent may do alone; approval governs the rest. Approval should bind to one specific action and its arguments (this payment, this amount, this beneficiary) rather than to the agent's task in general, and it should be required wherever an action cannot be undone or leaves the firm. Singapore's Model AI Governance Framework for Agentic AI, launched in January 2026, makes such checkpoints part of keeping humans meaningfully accountable (IMDA 2026). Approval is a control only if the approver actually decides. Chapter 16's worked example showed that a ten-step agent at 95% per-step reliability sends about a third of its tasks to review; a reviewer facing that volume who approves nearly everything is a rubber stamp, so approval and override rates are themselves monitored, as this section monitors a model's score distribution.

Change management. An agent is a composition (a model, its prompts, tools, memory, and permissions), and a change to any one is a change to the system. Vendors revise and retire hosted models on their own schedule, and a new version can change behavior on tasks the old one handled reliably, so a governed agent pins its versions, reruns its evaluation suite before promoting any change, and records the result. Adding a tool deserves particular care, because it changes what the agent can do, not just how well: it is a change in authority and should pass through the same approval as a change to a human's entitlements. The Upstart vignette that follows shows the same tension in a classical model: a supervisory assurance covers the system as reviewed, and a fast-changing system sits uneasily inside it.

Third parties. Most agents rest on a foundation model from an outside vendor and reach data through connectors, increasingly built on the Model Context Protocol Chapter 16 described, that may themselves come from third parties. Each is a third-party relationship in the sense of the U.S. banking agencies' 2023 interagency guidance, with its life cycle of due diligence, contract, ongoing monitoring, and exit (FRB, FDIC, and OCC 2023): the contract should give the bank notice of model changes, limits on the vendor's use of its data, and rights to the information it needs to validate. A connector is also an attack surface, since everything it returns enters the agent's context, so vetting connectors defends against the prompt injection Chapter 16 described and is more than a procurement step.

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How existing frameworks apply. As of October 2026, no framework a bank answers to was written for agents, and the existing ones reach them unevenly. In the United States, SR 26-2 (Federal Reserve, OCC, and FDIC 2026), which replaced SR 11-7 in April 2026, expressly leaves generative and agentic AI outside its scope pending separate treatment. That leaves no dedicated U.S. model-risk guidance for agents, but it does not leave agents ungoverned: a classical model an agent calls, such as a credit score or a fraud model, remains inside SR 26-2; the agent itself falls under operational-risk, third-party, and conduct expectations; and many banks extend their model-risk programs to agents voluntarily. The EU AI Act (EU 2024) has no agent category. It classifies by use, so an agent that scores the creditworthiness of natural persons is a high-risk system with the logging and human-oversight obligations above, applicable from December 2027 under the Digital Omnibus (Table 18.4), while the general-purpose model underneath carries its provider's own obligations, in force since August 2025. The NIST AI Risk Management Framework (NIST 2023), with its Generative AI Profile of July 2024 (NIST 2024), supplies the govern-map-measure-manage structure, and in February 2026 NIST's Center for AI Standards and Innovation launched an AI Agent Standards Initiative on agent security, identity, and interoperability (NIST 2026). ISO/IEC 42001, published in December 2023 (ISO/IEC 2023), is a certifiable management-system standard that covers agents as it covers any AI system, by requiring an organization to inventory, assess, and control them, though it prescribes no agent-specific controls. The most agent-specific texts are supervisory and voluntary: FINRA's 2026 report and Singapore's framework, both cited above. Table 18.7 collects the controls this subsection argues for, with the framework provision each most directly answers.

Control	What it requires	Framework hook (as of October 2026)
Mandate and owner	Written scope of tasks, tools, data, counterparties, and value thresholds; a named accountable human	FINRA 2026 (scope and authority); IMDA (human accountability)
Agent identity and least privilege	Own service credentials; entitlements reviewed like a person's; no accumulated conflicting entitlements	Segregation of duties (Chapter 10); IMDA (bound risks upfront)
Approval of irreversible actions	Approval bound to one action and its arguments; approval and override rates monitored	EU AI Act Art. 14; IMDA checkpoints
Action log	Instruction, retrieved context, every tool call and result, approvals, versions in force	EU AI Act Art. 12; FINRA 2026 (track actions)
Limits and kill switch	Caps on value per action, actions per hour, and spending; halt to a safe state	EU AI Act Art. 14(4)(e)
Pre-deployment evaluation	Task suite scored on end state; trajectory review; prompt-injection tests	NIST AI RMF (measure); SR 26-2 by analogy
Change control	Pinned model, prompt, and tool versions; evaluation rerun before promotion; new tools approved as new authority	ISO/IEC 42001; SR 26-2 by analogy
Vendor and connector oversight	Change notification, data-use limits, audit rights; connectors vetted	2023 interagency third-party guidance
Ongoing monitoring	Task success, escalation, override, and cost rates; drift in inputs	SR 26-2 by analogy; NIST AI RMF (manage)

Table 18.7: A control checklist for AI agents in a regulated financial institution. "By analogy" marks guidance that excludes agentic AI but whose discipline banks commonly extend to it.

None of these controls is new. Each is the discipline a bank already applies to a human employee, a trading algorithm, or an outsourced service, written down for an actor that combines features of all three. What is new is the order: for a scoring model, monitoring can follow deployment, because a human still stands between the score and its consequence; for an agent, the mandate, limits, log, and kill switch must exist before its first action, because afterward there may be nothing left to intercept.

18.5.3 Case Vignette: Regulatory Sandboxes and the Upstart No-Action Letter

Section 18.4.3 showed a regulator responding to a public controversy after deployment; the CFPB's engagement with Upstart Network shows the same concerns, whether an AI underwriting model produces disparate impact and can be adequately explained, addressed in advance. In 2017 the Consumer Financial Protection Bureau granted Upstart a no-action letter, a formal assurance that the agency would not bring

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an enforcement action against specific practices. Upstart is an online lender that uses machine learning for credit underwriting, including alternative data such as educational attainment. The letter was conditioned on Upstart regularly reporting detailed fair-lending test results to the CFPB, including disparate-impact statistics like Section 18.4.2's adverse impact ratio, broken out by race, ethnicity, and sex.

The CFPB issued a renewed letter in November 2020, which also required Upstart to notify the agency of significant model changes before making them. That condition ended the arrangement. In April 2022 Upstart told the CFPB it intended to add a significant number of new variables to its underwriting model; the Bureau asked for more time to evaluate the changes; and Upstart, citing the need to keep its models accurate during a period of rapid economic change, asked for the letter to be terminated, which the CFPB did by order in June 2022. The ending is itself a lesson in monitoring: a supervisory assurance covers the model as reviewed, and a model that must be retrained quickly sits uneasily inside it. The contrast with the Apple Card case is instructive. Upstart's reporting reportedly showed its model approving a meaningfully higher share of applicants, at similar or lower average interest rates, than a traditional credit-score-only model would have, evidence that a more flexible model can *expand* credit access rather than merely risk disparate impact. That evidence was credible, and available to the regulator at all, only because Upstart had committed in advance to the ongoing fairness testing and reporting that Sections 18.4.2 and 18.5.1 develop. A regulatory sandbox or no-action letter does not waive the underlying obligations (disparate-impact testing, adverse-action explanation, ongoing monitoring); it is a supervised space in which a firm demonstrates that a novel model meets them before, rather than only after, a wide public rollout.

18.6 Advanced Topics

Four further topics extend the chapter's toolkit: making a counterfactual explanation actionable rather than merely valid; a causal rather than statistical definition of fairness; extending explanation methods to the deep architectures of Chapters 16 and 17; and attaching a finite-sample uncertainty guarantee to individual predictions.

18.6.1 Algorithmic Recourse: Making Counterfactuals Actionable

Section 18.3.4's counterfactual, reduce debt-to-income by about four percentage points, is only one way to cross the approval threshold. Holding debt-to-income at the applicant's actual 28% and solving for credit history instead gives approximately 13.0 years, an increase of about 9.0 years from the actual 4. Both solve the same equation, $z \leq -0.330$, but they are not remotely equally achievable: an applicant can pay down debt over months, while nobody can shorten the wait for nine more years of credit history.

Algorithmic recourse (Ustun et al. 2019) formalizes this by attaching a cost to changing each feature and recommending the counterfactual with the lowest total cost rather than the nearest one in raw feature space. With debt-to-income costing 1 unit per percentage point (roughly, the difficulty of paying down that much debt) and credit history 5 units per year (it can only be waited out), the debt-to-income path costs 3.95 and the credit-history path $9.04 \times 5 = 45.19$, quantifying what the raw numbers already suggest: debt-to-income is the only path a lender should recommend. The lesson generalizes. A production counterfactual system should distinguish mutable features an individual can act on, such as debt-to-income, savings, or a co-signer, from effectively immutable ones: age, the passage of time, and any feature correlated with a protected characteristic. Recommending recourse through an immutable feature is unhelpful, and it can raise the fair-lending concerns of Section 18.4.2.

Recourse is a causal question. A counterfactual explanation is a statement about the model: move debt-to-income to 24.05% and this scorer approves. Recourse is advice to act in the world, and an action is an intervention in the sense of Chapter 6's Section 6.4.6. Actions rarely move one feature alone. An applicant who pays down debt out of savings lowers debt-to-income but also the savings balance a richer model would score; one who takes a second job raises income but may cut short their tenure with their current employer. The nearest counterfactual treats features as independent dials, while real actions move several at once through causal links, so a recommendation computed as if one feature moved can fail to deliver approval once the knock-on changes are scored. Recourse is therefore best framed as the cheapest intervention on a causal model of the applicant's features rather than the nearest point across the decision boundary (Karimi et al. 2021). A second gap is subtler: crossing the threshold need not change the applicant's true default

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risk. Chapter 6's credit-limit example showed a feature, the limit-increase flag, that predicts lower default while causally raising it; recourse through a feature of that kind, such as opening a new account purely to lengthen recorded credit history, earns approval without earning creditworthiness. The loss falls on the lender and, repeated across many applicants, becomes the drift Section 18.5.1 monitors. Recommending actions that move a real cause of repayment, such as paying down debt from income, keeps approval and actual risk aligned.

18.6.2 Counterfactual Fairness: A Causal Perspective

Section 18.4.2's disparate-impact and equalized-odds definitions are purely statistical: they compare outcome or error rates across groups with no model of *why* the rates differ. **Counterfactual fairness** (Kusner et al. 2017) asks a causal question instead: would this individual's decision have been different had their protected attribute (sex, race) been different, holding fixed everything about them that is not causally downstream of that attribute?

This reframing picks up where the Apple Card vignette (Section 18.4.3) left off. A joint account's credit history can be causally downstream of a couple's gender-influenced decision about whose name assets are held under, so a model using joint-account history is not necessarily counterfactually fair even though it never reads sex or gender. Holding the actual, gender-influenced account history fixed while imagining a different gender is the counterfactual comparison this definition requires, and the correlation-only reasoning of Section 18.4.2 cannot express it. Operationalizing it requires a causal graph specifying which features are downstream of the protected attribute, a much harder modeling commitment than any statistical metric, which is why counterfactual fairness is more common as a lens for diagnosing cases like Apple Card's after the fact than as a routine pre-launch test.

The definition is Chapter 6's Section 6.4.6 applied to a protected attribute. Treat the attribute as the "treatment" in the potential-outcomes framework, and counterfactual fairness asks that this applicant's decision be the same under both potential values of it; the causal graph plays the role that knowing the confounder Z played in that section's credit-limit example. Two lessons carry over. First, dropping the protected attribute is the analogue of the short regression: its influence still flows through correlated features, joint-account history or a ZIP code, as creditworthiness flowed into the limit-increase flag, and the omitted-variable formula describes how a variable left out of a regression leaves its effect behind in the variables kept in. Second, that section's noisy-proxy estimate, which recovered only $+0.0027$ of a true $+0.02$ effect, warns that adjusting for an imperfectly measured variable removes only part of a bias: a fairness repair that conditions on a coarse proxy for the legitimate driver of risk can leave much of the protected attribute's influence in place while appearing to have removed it. One thing does not carry over: there is no randomized experiment to fall back on, since a protected attribute cannot be assigned at random. Counterfactual fairness therefore rests on an assumed causal graph, which must be argued for and documented because it cannot be tested.

18.6.3 Explaining Deep Models: Saliency, Integrated Gradients, and the Faithfulness Problem

Section 18.3.5's LIME surrogate treats the network as a black box; when the model's own gradients are available, as they are for any network trained by backpropagation, a more direct family of methods computes an attribution straight from those gradients rather than from a sampled local regression. The simplest version, a *saliency map*, is the gradient of the output with respect to each input, the analytical local gradient Section 18.3.5 already computed, $(0.0469, -0.0666)$ at x_0.

Multiplying this gradient by the full distance to a baseline, $x_0 - x' = (1.0, 0.5)$ from $x' = (0, 0)$, gives a naive attribution of $(0.047, -0.033)$, summing to 0.014, but the network's output changes by $0.558 - 0.532 = 0.026$ between these two points, nearly double the naive estimate. The discrepancy is not rounding. The straight-line path from x' to x_0 crosses a kink: the second hidden unit's ReLU is inactive at x' (pre-activation -0.1) and switches on at $t = 0.4$ of the way along, so a gradient evaluated only at the endpoint x_0 cannot see the nonlinearity the path passes through earlier.

Integrated gradients (Sundararajan et al. 2017) fixes this by integrating the gradient along the entire path rather than evaluating it once,

$$\mathrm{IG}_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F}{\partial x_i}\big(x' + t(x - x')\big)\, dt. \tag{18.5}$$

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Because this network's pre-activations are piecewise linear in t along the straight path, the integral has an exact closed form on each of the two segments (before and after the ReLU switches on at $t = 0.4$), using the fact that $\sigma'(a + bt)$ integrates to $\sigma(a + bt)/b$. Carrying this out gives $\mathrm{IG}_1 = 0.0520$ and $\mathrm{IG}_2 = -0.0260$, summing to 0.0260, which matches the true output change up to rounding: the *completeness* axiom integrated gradients is designed to satisfy and the endpoint-gradient shortcut failed. This matters for the deep architectures of Chapters 16 and 17: CNNs and attention-based transformers admit the same gradient-based treatment, and integrated gradients (or a close variant) is the standard way to produce a saliency map for an image classifier or a token-level attribution for a language model.

A subtler warning concerns attention weights, since Chapter 16 introduced self-attention and it is tempting to read a high attention weight as "the model relied on this token." Empirical work (Jain and Wallace 2019) has shown that attention weights often fail a basic test of any explanation: alternative attention distributions can be found that leave a transformer's output essentially unchanged, so the observed weights are not uniquely responsible for the prediction in the way an explanation implies.

The same faithfulness concern extends to a large language model's own natural-language, chain-of-thought explanation of its reasoning. Research on LLM-generated rationales (Turpin et al. 2023) has found cases where a model's stated reasoning does not match the factor that changed its answer: the explanation is fluent and plausible but not faithful, in this chapter's technical sense, to the computation that produced the output. Gradient-based methods like integrated gradients, computed from the model's own mathematics rather than from a separately generated explanation, therefore remain the more trustworthy default for a high-stakes financial deployment of Chapter 16 or Chapter 17's architectures, however fluent a model's self-reported explanation may read.

18.6.4 Conformal Prediction: Honest Uncertainty for Individual Decisions

Every explanation method in this chapter says why the model predicted what it did; none says how *sure* the model is entitled to be. Chapter 6's Section 6.4.5 introduced split conformal prediction for a regression: hold out a calibration set, take the $\lceil (n+1)(1-\alpha) \rceil$-th smallest of its nonconformity scores, and wrap any model's point forecast in an interval whose coverage holds in finite samples, provided the data are exchangeable. Two things change when the model is a credit classifier deciding one applicant at a time.

First, the output becomes a set of labels. With one minus the probability the model assigns to the true label as the nonconformity score, and $\hat{q}$ the same calibration quantile, an applicant with predicted default probability $\hat{p}$ has a prediction set containing "repay" if $\hat{p} \le \hat{q}$ and "default" if $1 - \hat{p} \le \hat{q}$. A singleton set is a decision the model can stand behind at the chosen coverage; the set {repay, default} marks a case it cannot resolve, a natural trigger for the human review that Section 18.5.1's governance framework provides for.

Second, the guarantee Chapter 6 proved is marginal, an average over all applicants, and an average can hide a group the model serves badly. The companion notebook simulates an applicant pool in which 20% have thin credit files, whose default risk depends mostly on information the lender does not observe, fits a logistic regression on a score and a thin-file indicator, and calibrates on 2,000 applicants, 376 of them thin-file, at $\alpha = 0.10$. Pooled calibration gives $\hat{q} = 0.7025$. Averaged over 2,000 fresh calibration sets, its sets cover 90.07% of all applicants, as promised, but 91.75% of thick-file applicants and only 83.43% of thin-file ones: the guarantee holds on average while failing the group the model knows least about. *Group-conditional* (or Mondrian) calibration computes a separate quantile within each group, here 0.6903 from the 1,624 thick-file applicants and 0.7795 from the 376 thin-file ones, restoring 90.17% and 90.11% coverage. The repair has a visible price: the share of thin-file applicants whose set is {repay, default}, and who are therefore referred to a human, rises from 40.0% to 60.9%, while the thick-file share falls from 21.5% to 16.6%. That is the honest answer: the model cannot decide most thin-file cases at 90% confidence, and pooled calibration had hidden this behind overconfident singletons.

Three cautions apply. Group-conditional calibration needs the group at decision time, and calibrating separately by a protected attribute would use that attribute in the decision, raising the disparate-treatment concerns of Section 18.4.2; in practice a lender calibrates on legitimate operating segments, such as file thickness, product, or channel, and *audits* coverage by protected group in the spirit of that section's tests. Each group also needs enough calibration observations, since a quantile estimated from 376 cases varies more from one calibration set to the next than one estimated from 1,624. And the guarantee lasts only as long as the calibration data stay relevant: under the drift that Section 18.5.1's population stability index

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monitors, yesterday's quantile silently loses today's coverage, as Chapter 6's stress-regime example showed for a regression interval.

18.7 Challenges in Explainability, Fairness, and Regulation

Several challenges recur across this chapter's methods and connect directly to themes from earlier chapters.

Explanation methods are themselves approximations, not ground truth. Section 18.3.5's worked example showed a local surrogate matching the true model almost exactly, and showed how that agreement degrades once a perturbation crosses a kink in the underlying function. A post-hoc explanation is only as trustworthy as the neighborhood or reference baseline it was built around, a modeling choice with no universally correct answer.

Multiple fairness definitions cannot generally be satisfied simultaneously. Section 18.4.2's worked example, and the impossibility results underlying it, mean that a bank's choice of fairness criterion is an explicit policy decision with real distributional consequences, not a detail a sufficiently sophisticated model can sidestep.

Proxy discrimination survives the removal of protected characteristics. Section 18.4.3's case vignette showed that omitting a protected characteristic from a model's inputs does not guarantee outcomes free of disparate impact along that characteristic, since facially neutral features can correlate with it closely enough to reproduce the pattern.

Explainability and model capability can be in tension. Chapters 6, 16, and 17 each observed that some of the most powerful models (deep ensembles, attention-based transformers, convolutional and recurrent networks) are among the least interpretable. Post-hoc methods manage that tension rather than eliminate it, which is why some regulated uses, consumer credit decisions in particular, still favor interpretable models over marginally more accurate black boxes.

Monitoring must anticipate adaptive, not just random, drift. Sections 18.4.4 and 18.5.1 show that in an adversarial financial setting the population being scored can learn from and react to the model itself, so drift detection is an ongoing arms race rather than a one-time validation exercise.

Governance frameworks require organizational capability, not just documentation. Section 18.4.3 showed that a bank can have a governance framework on paper and still fail the underlying obligation, producing no usable explanation when a customer or regulator asked for one, which is why SR 11-7 and the NIST AI RMF both emphasize ongoing practice over a one-time compliance checklist.

18.8 Key Takeaways

Interpretability is a property of a model's structure; explainability is a property of a method applied to a model. The distinction matters because most of this book's powerful models, Chapter 6's ensembles, Chapter 16's transformers, and Chapter 17's CNNs and RNNs, are not interpretable and depend on the post-hoc methods of Section 18.2.1's taxonomy. Global methods (Section 18.3.1) describe a model's behavior in general, and Chapter 15's fraud model showed that a feature can matter in principle yet show zero measured importance on a specific sample. Local methods answer a single case's question. An exact Shapley decomposition (Section 18.3.2) of a model linear in the log-odds, like Chapter 9's credit scorer, yields adverse-action reason codes and a counterfactual explanation almost for free. LIME (Section 18.3.5) extends the idea to nonlinear models such as Chapter 6's toy network, with validity guaranteed only in a local neighborhood whose size is itself a modeling choice.

Fairness (Section 18.4.2) is not a single statistical target: the four-fifths rule and error-rate parity can point in different directions on the same model and data, as the two-group audit showed, and theory proves the tension cannot generally be resolved away. The 2019 Apple Card controversy (Section 18.4.3) showed a widely used consumer credit product failing the fairness-testing and explanation obligations this chapter formalizes without ever using a protected characteristic as an input.

Section 18.4.4 extended Chapter 15's structuring vocabulary from evading a reporting threshold to evading a model, and Section 18.5.1 showed that a leading indicator like the population stability index can flag a shift in the scored population before enough labeled outcomes accumulate for a lagging indicator like precision. Together they tie explainability, fairness, robustness, and governance into the single discipline that

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SR 11-7 and the NIST AI Risk Management Framework formalize. AI agents that act rather than score (Section 18.5.2) extend it from outputs to actions: a mandate and a named owner, least privilege, approval bound to each irreversible action, a complete action log, hard limits with a kill switch, and change control over model, prompts, and tools, all in place before the first action, at a time when SR 26-2 leaves agentic AI out of scope and the EU AI Act reaches agents only through their use.

Section 18.6 pushed each thread deeper: cost-weighted recourse showed that not every valid counterfactual path is achievable; integrated gradients showed where a naive gradient shortcut understates a deep model's attribution, and why attention weights and chain-of-thought rationales cannot always be trusted as faithful explanations of Chapter 16's models; and the CFPB's Upstart no-action letter (Section 18.5.3) showed the same obligations met proactively, through structured regulatory engagement, rather than after a public controversy like Apple Card's.

18.9 Suggested Exercises

18.1 Using Table 18.1's method, permute the late-night indicator using a different fixed rearrangement (a cyclic shift by one position rather than a full reversal) and recompute the confusion matrix. Does the conclusion that late-night timing shows zero permutation importance on this sample still hold?

18.2 Explain why a feature can have a strong, monotonic partial-dependence curve (Table 18.2) while simultaneously showing zero permutation importance (Table 18.1) on the same dataset.

18.3 Using Section 18.3.2's method, compute the exact Shapley decomposition for a second applicant with a 32% debt-to-income ratio and 7 years of credit history, using the same reference baseline, and verify the efficiency property (that the two contributions plus the baseline log-odds sum exactly to this applicant's own log-odds).

18.4 Rank the two contributions from Exercise 18.3 and state which reason code would appear first on this applicant's adverse action notice.

18.5 Using Section 18.3.4's method, compute the counterfactual credit history (holding debt-to-income fixed at 28%) that would have gotten Chapter 9's original applicant approved at the 620-score threshold, and compare the required change to the required debt-to-income change computed in the text.

18.6 Using Table 18.3's network, compute the network's exact prediction at $x = (1.05, 0.55)$ and compare it to the LIME surrogate's prediction at the same point, reporting the discrepancy.

18.7 Explain why the LIME surrogate fitted in Section 18.3.5 recovers the analytical local gradient almost exactly, referencing which structural feature of the network (the sign of each hidden unit's pre-activation) makes this possible.

18.8 Using Section 18.4.2's formula, compute the adverse impact ratio if Group Beta's approval rate in Table 18.5 were instead $2/4 = 50\%$ (holding Group Alpha's rate fixed), and state whether the four-fifths rule would still be violated.

18.9 Using Table 18.5, compute each group's false-positive rate (the fraction of applicants who will not actually default but are nonetheless denied), and discuss whether this statistic points in the same direction as the false-negative-rate comparison in the text.

18.10 Explain, using the impossibility result cited in Section 18.4.2, why a lender cannot generally fix Table 18.5's adverse impact ratio and its false-negative-rate gap with a single adjustment to the model's classification threshold alone.

18.11 Using Section 18.4.3's case vignette, explain why removing sex or gender as a model input does not, by itself, guarantee that a credit-limit algorithm's outcomes are free of gender-correlated disparate impact.

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18.12 Using Section 18.4.4's method, compute the amount z-score reduction needed to flip Chapter 15's transaction 6 (amount z-score 3.4, distance 9.2, late-night flag 1) from fraud to not-fraud, holding distance and late-night status fixed, and compare the required reduction to transaction 1's.

18.13 Explain why the fraud-scoring model in Section 18.4.4 is harder to game with a single-feature perturbation than a deep image classifier is with a pixel-level perturbation, referencing the number of independently suspicious features each transaction carries.

18.14 Using Table 18.6's formula, compute the total PSI if the current-period distribution instead matched the baseline distribution exactly, and explain what this result confirms about the PSI formula's behavior when there is no drift at all.

18.15 Using the same PSI formula, compute the total PSI for a more extreme drift scenario in which the current distribution is $(0.20, 0.25, 0.30, 0.25)$ (holding the baseline fixed), and state whether this would fall in the "no material change," "moderate shift," or "major shift" band described in the text.

18.16 Explain why the population stability index is described as a leading indicator and Chapter 15's precision-drift table as a lagging indicator, and describe one practical consequence of a model governance program monitoring only the lagging one.

18.17 Using Section 18.5.1's description of SR 11-7 and the NIST AI RMF, map each of this chapter's twelve worked examples to the framework function (govern, map, measure, manage) it most directly supports, and justify any example you assign to more than one function.

18.18 Compare Section 18.3.2's exact, closed-form Shapley decomposition to Section 18.3.5's sampled, approximate LIME surrogate, and explain the specific structural property of the credit-scoring model (as opposed to the toy neural network) that makes the closed form available in the first place.

18.19 Using Section 18.6's method, compute the algorithmic-recourse cost of a third counterfactual path that changes both debt-to-income and credit history by equal fractions of their individually required changes (that is, half of the debt-to-income path and half of the credit-history path combined), and compare its total weighted cost to the two single-feature paths in the text.

18.20 Using Section 18.6's integrated-gradients derivation, recompute IG_1 and IG_2 using a baseline of $x' = (0.5, 0.25)$ instead of $(0, 0)$, and verify that the two attributions still sum to $\hat{y}(x_0) - \hat{y}(x')$ at this new baseline, illustrating that such an attribution is defined only relative to a chosen reference point.

18.21 Using Section 18.4.2's calibration-in-the-large check, recompute each group's average predicted default probability if Alpha-3's predicted probability were instead 60% rather than 95.5% (holding every other applicant's prediction fixed), and state whether Group Alpha's calibration gap (average predicted probability versus its 50% actual default rate) widens or narrows as a result.

18.22 Explain why the fact that Group Alpha and Group Beta share an identical 50% actual default rate in Table 18.5 does not, by itself, guarantee that a model calibrated on the combined population will also be equally calibrated within each group separately.

18.23 A bank plans an agent that reconciles failed trades: it reads the trade and settlement records, identifies the cause of each break, drafts a correcting entry, and books entries below $10,000 itself. Using Section 18.5.2 and Table 18.7, write the agent's mandate (tasks, tools, data, value thresholds, owner), classify each of its tools as read-only, reversible, or irreversible, and specify which actions require human approval and what the action log must record for an auditor to reconstruct a booked entry a year later.

18.24 The vendor that supplies the foundation model under the agent in the previous exercise announces that the model version the bank uses will be retired in ninety days. Using Section 18.5.2's discussion of change management and third parties, describe the steps the bank should take before switching to the new version, and explain why SR 26-2's exclusion of agentic AI does not relieve it of doing so.

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18.25 Using Section 18.6.4's rule, find the range of predicted default probabilities $\hat{p}$ for which the prediction set is {repay}, {default}, or {repay, default} under the pooled quantile $\hat{q} = 0.7025$, and again under the thin-file quantile $\hat{q} = 0.7795$. Explain why the wider thin-file quantile raises the share of thin-file applicants referred to human review, and why no applicant's set can be empty when $\hat{q} \geq 0.5$.

18.26 Chapter 6's Section 6.4.5 showed that, given one calibration set, coverage follows a Beta(k, $n+1-k$) distribution. (a) Using Section 18.6.4's group sizes, n = 1,624 thick-file and n = 376 thin-file calibration applicants at $\alpha = 0.10$, compute k for each group and the mean and standard deviation of the corresponding Beta distribution (a Beta(a, b) variable has mean $a/(a+b)$ and variance $ab/[(a+b)^2(a+b+1)]$). (b) Explain what the difference in standard deviations means for a validator who checks thin-file coverage on a single calibration set. (c) Explain why a lender should not instead calibrate separately by race or sex, and what it should do to check coverage across those groups.

18.27 **Challenge (real data).** Download a recent year of HMDA (Home Mortgage Disclosure Act) loan application register data for a single large lender (freely available from the Consumer Financial Protection Bureau's HMDA data browser, no login required). (a) Using Section 18.4.2's adverse impact ratio formula, compute the approval-rate ratio between two applicant groups of your choosing (for example, by race or ethnicity, both reported in the data), and state whether the four-fifths rule is violated, as it was in Table 18.5's toy example. (b) Fit a simple logistic regression predicting loan approval from the legitimate underwriting variables HMDA reports (income, loan amount, debt-to-income category), and recompute the adverse impact ratio using the model's predictions instead of the raw historical approval decisions, discussing whether the disparity narrows, widens, or stays about the same. (c) HMDA data reports the reason lenders gave for each denial; compare the stated denial reasons for the two groups you chose, and discuss whether this real, lender-reported information changes your interpretation of any disparity found in part (a), relative to the toy example in the text, where no such reason codes were available to explain the underlying gap.

18.10 Further Reading

- Lundberg and Lee (2017), "A Unified Approach to Interpreting Model Predictions." The original SHAP paper unifying Shapley-value-based feature attribution, underlying Section 18.3.2.
- Ribeiro et al. (2016), ""Why Should I Trust You?": Explaining the Predictions of Any Classifier." The original LIME paper underlying Section 18.3.5.
- Shapley (1953), "A Value for n-Person Games." The original cooperative-game-theory paper defining the Shapley value that Section 18.3.2 applies to a single credit decision.
- Molnar (2022), *Interpretable Machine Learning*. A comprehensive, freely available treatment of permutation importance, partial dependence, Shapley values, and LIME, covering the full taxonomy in Figure 18.1.
- Doshi-Velez and Kim (2017), *Towards a Rigorous Science of Interpretable Machine Learning*. A foundational discussion of what interpretability means and how to evaluate it rigorously, underlying the distinction drawn in Section 18.2.1.
- Barocas et al. (2023), *Fairness and Machine Learning*. A comprehensive treatment of statistical fairness definitions, including disparate impact and equalized odds, underlying Section 18.4.2.
- Chouldechova (2017), "Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments." One of the two papers establishing the impossibility of simultaneously satisfying calibration and error-rate parity across groups, cited in Section 18.4.2.
- Kleinberg et al. (2017), "Inherent Trade-Offs in the Fair Determination of Risk Scores." The companion impossibility result to Chouldechova (2017), cited in Section 18.4.2.
- Dwork et al. (2012), "Fairness Through Awareness." The original individual-fairness definition discussed as a third fairness paradigm in Section 18.4.2.

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- Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency (2011), *Supervisory Guidance on Model Risk Management*. The foundational U.S. bank regulatory guidance on model risk management underlying Section 18.5.1.
- Federal Reserve, OCC, and FDIC (2026), *Revised Guidance on Model Risk Management*. The April 2026 interagency guidance that superseded SR 11-7, discussed in Section 18.4.1.
- EU (2024), *Regulation (EU) 2024/1689 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)*. The EU regulation whose high-risk obligations for credit scoring and insurance pricing are summarized in Table 18.4.
- NIST (2023), *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*. The govern/map/measure/manage framework discussed at the close of Section 18.5.1.
- FINRA (2025), *2026 FINRA Annual Regulatory Oversight Report*. The first U.S. supervisory report to treat AI agents directly, listing the autonomy, scope-and-authority, and auditability risks discussed in Section 18.5.2.
- IMDA (2026), *Model AI Governance Framework for Agentic AI*. Singapore's voluntary governance framework for agentic AI, organized around bounding risks, human accountability, technical controls, and end-user responsibility, discussed in Section 18.5.2.
- Goodfellow et al. (2015), "Explaining and Harnessing Adversarial Examples." The foundational paper on adversarial perturbations in deep learning, contrasted with the fraud-model perturbation in Section 18.4.4.
- NYDFS (2021), *Report on Apple Card Investigation*. The regulatory report underlying the case vignette in Section 18.4.3.
- Ustun et al. (2019), "Actionable Recourse in Linear Classification." The foundational algorithmic-recourse paper underlying the cost-weighted counterfactual comparison in Section 18.6.
- Karimi et al. (2021), "Algorithmic Recourse: From Counterfactual Explanations to Interventions." The argument that recourse should be computed as a minimal intervention on a causal model of the applicant's features rather than as the nearest counterfactual, behind the causal discussion in Section 18.6.1.
- Kusner et al. (2017), "Counterfactual Fairness." The original causal counterfactual-fairness definition discussed in Section 18.6.
- Sundararajan et al. (2017), "Axiomatic Attribution for Deep Networks." The original integrated-gradients paper underlying Section 18.6's worked example on Chapter 6's network.
- Jain and Wallace (2019), "Attention is not Explanation." The empirical case against reading Chapter 16's attention weights directly as explanations, discussed in Section 18.6.
- Turpin et al. (2023), "Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting." The empirical study of unfaithful LLM-generated rationales discussed in Section 18.6.
- Angelopoulos and Bates (2023), "Conformal Prediction: A Gentle Introduction." A practical introduction to conformal prediction, including prediction sets for classifiers and group-conditional coverage, extending Chapter 6's split-conformal method to Section 18.6.4's credit decisions.
- CFPB (2017), *CFPB Announces First No-Action Letter to Upstart Network*. The regulatory announcement underlying the case vignette in Section 18.5.3.

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